Sydney Sidetracks: Listening in to 'history where it happened' using the ABC's television and radio archives

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Sarah Barns is a strategist, researcher and producer with many years experience working on strategic policy and development issues related to interactive media and cultural services. In 2008 Sarah worked with the ABC to establish a new, multi-platform pilot initiative called Sydney Sidetracks, which showcases archival material capturing Sydney's changing street life from the collections of ABC's television and radio archives, as well as the National Film and Sound Archive (NFSA), the Powerhouse Museum, the State Library of NSW and others. Sydney Sidetracks is the result of her doctoral research currently being undertaken through the University of Technology, Sydney (UTS).

Abstract

Developments in mobile computing, including the increasing proliferation of 3G mobile devices with location-aware capabilities, have generated heightened interest among media practitioners, researchers and artists in the use of urban or environmental contexts as a media platform. The ABC's Sydney Sidetracks was the ABC's first foray into location-based services, and offers users an experience of 'history where it happened' out on the streets of central Sydney. This paper discusses the research methodology adopted in the development of Sydney Sidetracks, which has deliberately sought out ambient sound recordings capturing events and locations no longer visible today – constituting an archaeology of Sydney's invisible topography which locates its recorded action, not surviving artefact. The paper explores the opportunities associated with the use of sound archives on mobile platforms, where offering users an ability to 'listen in' to historical recordings in-situ might promote a different kind of interactivity, one that uses the sounds of another era to frame a contemporary visual experience of place. In this way the mobile listening device might be imagined as a kind of homing device to the recorded history of a place, which, rather than distancing the listener from their physical or social context, might– somewhat playfully – promote interaction with previous listening publics and former uses of public urban space. This approach is contrasted with what Crang and Graham (2007) have described as the more anticipatory focus of many location-based media interventions, which link imaginations and anticipations of future behaviour(s) to categorical renderings from computerized memory. An 'auditory turn' towards the sound archive is thus presented as an alternative methodology through which to explore the experience and representation of urban spatiality using networked media devices.

Keywords

radio archives, mobile media, urban history, sound history, Sydney, the ABC.

Introduction

It was 9am Sydney time on August 15 1945 when British Prime Minister Clement Atlee announced that Japan had surrendered unconditionally to the Allies. "The last of our enemies is laid low" he said. “PEACE!” roared *The Sun*. In Martin Place massive crowds spontaneously gathered to yell and dance in celebration of the end of the War. The Australian Broadcasting Corporation’s (ABC) Talbot Duckmanton was there that day to record the scenes, describing to his audience some of the finer details of the setting – a Hitler effigy being hung from the windows of one of the banks, circles of dancing women before him (“lovely looking ladies, too!”), the din of a mosquito zooming around maniacally over head.

The recording captures an emerging style of documentary radio reportage adopted more widely by the ABC’s radio correspondents after the war, as they took advantage of new, more versatile recording technologies to head out of the studio and into the streets and backwaters of Australia (Inglis 1983, p.164). As Martin Thomas (2007) has discussed, a greater sense of immediacy was possible through the mobile, technologically-mediated spatiality of documentary radio, which used first-person narration in order to situate the recordings in a specific time and place. According to Thomas, “scripted commentary during this period frequently emphasised that we are out in the field with recorder and microphone – that you are listening to the sound of ‘real’ Australia and not a confection cooked up by actors and the sound effects man in a city studio. The result of this was a sense of immediacy” (2007, p.19).

Standing in Martin Place today, listening in on an iPod to the sounds captured by the ABC’s recording of these ebullient scenes some sixty-five years ago, one feels a giddy sense of time travel, being transported to that triumphant moment as it had been experienced *right here*. The sense of immediacy established by Duckmanton’s reportage draws today’s listener back in, to share with his radio audience of the day in imagining the scenes being described; “there’s no policeman directing traffic on Pitt St today” he tell us all. Today there are crowds of snappily-dressed workers to be seen, plenty of them with iPods, milling about in the pedestrian promenade flanked by some of Sydney’s proudest sandstone buildings baking in another hot summer afternoon. Watching all this, while listening to the sounds of this re-purposed radio recording through headphones, a different kind of audio-visual interaction is experienced, one that combines the immediacy of the visual present with a recording of its 1945 aural equivalent.

ABC archival recordings capturing specific events and locations around central Sydney can today be heard in this way, re-cast back onto the places they originally documented via a new multi-platform pilot launched by the ABC in November 2008. *Sydney Sidetracks* (http://www.abc.net.au/sidetracks) offers access to more than 50 'points of interest' (POIs) around central Sydney, each one presenting a selection of archival audio, video, and images relating to the locations featured. Material can either be viewed via a map online, or can be downloaded to mobile phone or iPod, thus enabling users to take the material with them while out and about in Sydney, where they can tune into 'history where it happened'.

As well as the VP Day recording, *Sidetracks* also offers access to a number of other ambient, location-based audio excerpts featuring the sounds of Sydney as captured by the ABC's television and radio reporters over the years. Many of these relate to times of public unrest, which attracted media attention and necessitated 'on the ground' media coverage. These include the Builders' Labourers' Federation (B.L.F.) Green Bans protests as they held up development work around the Rocks and Woolloomolloo during 1973-4; the Vietnam war protesters gathering outside the old Commonwealth Centre (now Chifley Square); and Prime Minister Robert Menzies being heckled by communists at the now-demolished Sydney Stadium in 1948. Also featured are short audio compilations or 'soundwalks' that contain collections of interviews and archival field recordings capturing changing streetscapes and locations. These include a recording of the auctioning and demolition of the Hotel Australia, the demolition of the Pyrmont Incinerator designed by Walter Burley-Griffin, and a collection of recordings about the decade-long construction of the Sydney Opera House at Bennelong Point, including the sounds of ships on Sydney harbour the day of its opening in 1973 by Queen Elizabeth II[1].

Along with audio, *Sydney Sidetracks* also features extensive visual material, including film footage shot from the front of a tram moving down George St in 1906 held by the National Film and Sound Archive (NFSA), ABC TV's *This Day Tonight* investigatory footage about Sydney's criminal underworld during the 1960s and 1970s, and images sourced from the collections of the Powerhouse Museum, the City of Sydney, the Dictionary of Sydney, the State Library of NSW and the Museum of Contemporary Arts (MCA).

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[1] Individual audio recordings discussed in this paper can be found at http://www.sitesandsounds.net.au or through ABC *Sydney Sidetracks* at http://www.abc.net.au/sidetracks which offers access to recordings online or through a mobile application which can be downloaded direct to the phone. Note that the mobile application does not make use of GPS technology but rather includes all content in the phone application itself, to save users any potentially unpleasant data charges.

It's the sound-based elements of *Sydney Sidetracks* that form the particular focus of this paper, as they constitute the outcome of my research to identify ambient, location-based sound archives relating to central Sydney. The research has been inspired as a particular response to mobile devices and their increasing use not only for social communication but also interaction with a given locality – what's known as 'location-based services' – via incorporation of GPS satellite receivers and maps. The research has sought to explore what the activation of broadcast archives – in particular sound archives – might mean in mobile listening contexts. Rather than promoting a screen-based mobile phone experience, it has rather used the built environment as a spatial context or platform from which to excavate its invisible history, through the act of listening in to its recorded documentation.

Therefore unlike the traditional historic tour, *Sidetracks* mostly visits events and places that are no longer present. The 'sidetrack' is a journey that charts locations and events that are mostly invisible to the naked eye. In order to facilitate this, ambient, location-specific sound recordings were required – those that captured the actual sounds of the street, as Duckmanton's VP Day recording does so well. A particular research motivation was in locating audio material relating to sites that had radically changed in some way, such that the listening experience would establish a certain displacement, exploiting the *acousmatic* qualities of recorded sound, where the original source of the recording is hidden or distanced from our experience of it (Helyer, 2007). Listening to a recorded event or lost site as it was originally documented in-situ could on the one hand affect displacement – being from another time and capturing what can no longer been seen – just as it revisited the event 'here' as it 'really happened'.

This use of ambient sound archives has deliberately sought to enable listeners to participate, in-situ, in recorded moments of the past. By revisiting the broadcast radio archive, the experience is one of revisiting the occasions of the past as they were originally experienced – eg as the *present* – rather than listening to recollections of a historical event or environment through contemporary commentary or oral history. In other words, this has been a search for the auditory trace of past happenings and places, rather than the memory of them.

As well as offering what one blogger described as "an intriguing twist on the Google mash-up genre", research toward *Sydney Sidetracks* is therefore also the outcome of my attempt to locate a place for sound archives in the urban environments of networked, mobile media devices. This research foregrounds the practice of listening in to the invisible, historical terrain as a kind of archeology of the documented streetscape – an archeology of recorded action rather than surviving artifact. This aims to promote an experiential exploration of some of the dynamics and transformative events that have shaped Sydney's cultural and built fabric during a period of radical geographical change. It follows Alec Morgan (2005) when he writes:

> It is all too easy to fall into the trap of believing that the cultural essence of Sydney lies embedded in its architecture. It's structures, buildings and monuments. I find this method of interpreting the past, this reliance on concrete and real estate, a faulty and unsound foundation upon which to build an understanding of the forces that shape the distinctiveness of the city...
>
> I sense that there is another city lying undiscovered beneath these bloated, familiar carcasses and that cultural interpretation by architecture is too impoverished to satisfy a secret desire to connect to something of Sydney's past that is more elusive, more sensual, than a pile of bricks and mortar.

This research has necessitated access to public archival resources for the purposes of re-use, and engaging broadcasters and cultural archiving organisations in the potential to develop and deliver audio-visual archival collections as they relate to specific places – in-situ. My listening practice has to date been supported by the National Film and Sound Archive (NFSA) and the ABC. Through a process of auditioning hundreds of radio, television and film recordings, I have kept my ear close to the speakers, so to speak, listening out for ambient sounds often uncatalogued within both television and radio broadcast features, and listening out for the street scenes themselves, rather than those talked about or remembered from the vantage point of a recording studio.

In this paper I will discuss this listening methodology as a research practice that positions mobile media somewhat differently to a range of other media projects and perspectives that use or engage with location-based technologies in urban contexts. The practice of listening to documentary audio archives in a way that re-casts 'what really happened' back onto the streets of Sydney is in particular discussed as a deliberate, practice-led response to what Crang and Graham (2007) have suggested is an emerging 'politics of visibility' in the use of situated mobile media. I should add that my emphasis in this particular paper is not so much on the process of developing site-specific audio using archival sources, nor on the challenges associated with designing an interface for sound archives using online or mobile platforms. The focus here is rather on what a listening practice that is tuned in to the sounds of the archives might offer by way of a critically-engaged response to debates about the impact of networked, mobile media on our knowledge and experience of urban environments. This aims to extend a framing of the mobile listening experience beyond that of a purely intimate, personal experience of spatial contexts as described by Bull (2007) which is seen to disconnect the listener from their environment, and to instead connect with what Boyer (1996) has described as the "city of collective memory".

The virtual – getting kinda physical

"Isn't it strange how history has been replaced by technology?" (From Jean Luc Godard's *Eloge de l'Amour*, quoted in Toop 2004, p.40).

The increasing take-up of 3G mobile phones and release of Apple's second-generation iPhone into the Australian marketplace in June 2007 mean use of mobiles increasingly integrates a variety of functions that extend from basic 'talk and text' to include navigation via maps, access to mobile content via search engines, listening to music via mp3 and radio, and user creation of content such as images and videos. One of the more hotly-anticipated trends associated with growth of 3G mobile media has been the development of location-based services (LBS), which make use of a mobile phone's in-built GPS receiver to deliver 'geo-web' services tailored for use in specific locations. Using online map interfaces such as Google Maps, users can look up information about local business and retailers, search for relevant services nearby, or blog photos directly as map 'mash-ups'. According to a special feature on Google's entry into the geo-mapping space featured in *Wired* (2007), such applications are "changing the way we see the world", providing tools that let anyone with an internet connection layer their own geographic obsessions on top of ever more detailed road maps and satellite images. "We're all geographers now" says *Wired*.

While the majority of the mainstream mobile service offerings in Australia today are focused largely on promoting access to local retail and tourist services – a kind of *Yellow Pages* for the phone – the implications of geo-aware mobile computing for the way spatial contexts are represented and experienced are enormous. Indeed, they represent a shift toward what is known as 'pervasive' or 'ubiquitous' computing, when 'post-desktop' distributed computational intelligence becomes more embedded with the spatial structures of everyday life.

The possibilities of a context-aware, post-desktop computing environment have been anticipated for some years now. Since the late 1980s, computer scientists and engineers have been researching ways of embedding computational intelligence into the built environment (Galloway, 2008). In 1991, *Scientific American* published an article by Mark Weiser titled "The Computer for the 21st Century" calling for computational devices to become 'invisible' and shifting the focus away from solitary, immersive, virtual environments to a socially integrated context, in a way that "takes into account the natural human environment and allows the computers themselves to vanish into the background." Such a shift has more recently been described by the International Telecommunications Union (ITU) an "Internet of things" (2005) where the 'users' of the Internet will be counted in billions and where humans may become the minority as generators and receivers of information. This reflects the way that Radio Frequency Identification (RFID) tags, sensors, actuators, robotics and nanotechnology are in the process of making processing power increasingly available in smaller and smaller packages so that networked computing dissolves into the material fabric of the world around us.

Within ubiquitous computing environments, the spatial context is not a passive backdrop but an active agent in the way daily life is experienced and organised. According to Crang and Graham (2007), "the spaces around us are now being continually forged and reforged in informational and communicative processes. It is a world where we not only think of cities but cities think of us, where the environment reflexively monitors our behaviour". Otherwise put, the "street as platform" (Hill, 2008) is starting to emerge, one that makes possible real-time, multi-scalar interactions between the hyper-local, invisible, computational intelligence embedded in everyday environments and geographically distributed informational networks.

As recognized by those such as Bull (2004; 2007), Castells (2007) and Heise (2008), such mobile, context-aware computing applications recalibrate our experience of space and place, reconfiguring local interactions through multi-scalar interactions with global, networked publics. Taking advantage of the tools of geographic information systems (GIS), including computerised data gathering, aerial photography, remote sensing and satellite tracking, digital mapmakers use increasingly sophisticated graphics technology to build flexible visual encodings of this multivariate spatiality. The advantage of distributed mobile technology also means this new geography can be mapped in real time; movement of people, products and services can be monitored and evaluated via the computational intelligence embedded within them in real time.

In this way locative artist Esther Polak uses GPS technology to track the movement of agricultural products like milk, as they make their way across different countries (see milkproject.net). The Intel-funded *Urban Atmospheres Group*, MIT's *SENSEable Cities Lab*, Queensland University of Technology's Marcus Foth (2009), Arup's Dan Hill, author of the widely-read cities blog cityofsound.com, and Adam Greenfield, now Head of Design Direction at Nokia, have each promoted the applications of mobile computing for the development of a mobile-enabled participatory urban planning & design culture, one that incorporates not only professional data collection but also the informational life-worlds of millions of mobile users, thus facilitating a more ‘bottom-up’, citizen-oriented planning process (see, for example Beyea, Geith & McKewon, 2008).

The use of hyper-local, multi-scalar and real time mapping techniques has been taken up as a means to expose ‘hidden’ or hitherto invisible relationships, including the relationships between centre and periphery, power and influence (see Sassen, 2008; Boyer 2006). Maps can, as one practitioner enthuses, “be studied and interpreted, and used to generate new metadata so we can reorganize otherwise undiscovered opportunities” (van Weelden 2006, p.26). The more network mapping can be used to describe the connectivity of the physical world, it is claimed, “the more we find we are actually studying versions of metadata – economic, political, cultural, religious – that describe how we intentionally and unintentionally run our physical world” (ibid).

Where the Euclidian framing of geographic space was subjected to sustained critique for its simplistic organisation of complex spatiality, the use of networked data as the “vector of a new geography” (Holmes, 2007) is today celebrated. As Peter Hall and Janet Abrams suggest “[m]apping has emerged in the information age as a means to make the complex accessible, the hidden visible, the [once] unmappable mappable” (2006, p.12).

Talking such tools out onto the street, a new guard of locative media practitioners have sought to engage with the ambience of the streetscape as a platform from which to construct alternate visions of the city. In their 2007 survey of ‘Sentient Cities’ Crang and Graham (2007) explore the way location-aware computing has greatly expanded the possibilities for the work of artists and architects to attempts to re-enchant the world, or to “offer a way of making visible all these hidden stories of place [...]Where once there were official and dominant memories inscribed on the city now these stories from below can be added” (2007, p.815). The soundwalks of Janet Cardiff overcode the present city with memories of the past, producing a space that is not quite of the now but is rather haunted by ghostly, technologically preserved or recalled presences (Pinder, 2005). The project *Murmur* (http://murmure.ca/) first launched in Toronto's Kensington Market in summer 2003, then Vancouver's Chinatown and then Montreal, allows people to take a web based map and tag sites with stories and memories of what the places mean to them in a form of 'intimate commemoration'. Many other projects have developed this theme, especially the possibility of collaborative authoring connected to locative media. The UK artist group Proboscis, for example, have developed Urban and Sonic Tapestries to encourage users to create their own recordings of their everyday habitats (see urbantapestries.net). In Sydney the UK arts group Blast Theory introduced their Ride Spoke game into the Rocks in 2009, offering participants GPS-equipped bikes to explore and document the psyche of the city and its inhabitants, as riders search for undiscovered hiding places and record their own responses to the terrain.

While location-specific arts and media practitioners seek to create new social spaces for interaction in the city, others question the impact of real time media in its effect on urban spatiality. The critique of much cartographic practice as a “science of siege” (Mattelart, 2002) has recognized that all mapping is bound within specific discursive frameworks, that is, it is always provisional. As the geographer Denis Cosgrove emphasised, urban space and cartographic space remain inseparable: “as each is transformed their relationship alters” (quoted in Hall and Abrams 2006, p.157). Put another way, maps do not simply represent geographies or mobile users; they effect their actualization. The activity of network mapping and locative media may be attuned to the relational and multi-layered nature of its known world, but it nevertheless cannot avoid the radically reorganizing implications of its own vision.

As Manuel Castells has argued, through its functional logic of real-time interaction, the network society exhibits a tendency to eviscerate the distinctive qualities of different places (2000, p.147). Physical space is converted into ‘spaces of flow’, connected as nodes on a network, which are able to flexibly interact within one another, no matter where they are located, because they are all ‘here’ in the same time-sharing environment that is the space of flows (see Stalder 2006, p.153). To Castells, the essentially placeless logic of informational networks occurs through the separation of place-based information from the geographic environment, the neighboring local population and their local cultures. This does not mean that spaces of flow are placeless, he argues, but that territorial configurations shift towards the relational structures of communication networks rather than territories.

To Castells, the diffusion of mobile communication greatly contributes to this reconfiguration of spatial experience, linking social practices in multiple places, and desequencing of social action through what he calls ‘timeless time’ (Castells 2007, 172). The always-on, always-accessible network produces a broad set of changes to our concept of place, linking specific locales to a global continuum and thereby transforming our sense of proximity and distance (Varnelis and Friedberg).

Thus, not unlike the visions of modernist urban planners and architects such as Patrick Geddes, Ebenezer Howard and LeCorbusier, whose abstracted plans for the city were used to progress utopian social agendas of the city (Hall 1988, p.3), the datascapes of today's urban modelers and geo-web practitioners can be seen to contain their own hidden epistemologies of power and abstraction within their geographic gaze: offering what Boyer has described as "a detached pictorialisation of the world, grasped by an elevated, disembodied, scanning eye" (Boyer 2006, p.328).

To Crang and Graham (2007), embedded within the transformed life-worlds of urban informatics and mobile media is a 'politics of visibility', one that emerges both in making the technologies visible to us and in how we are made visible to them (2007, p.789). The authors suggest an increasing saturation of urban spaces with anticipatory technologies profile users in more sophisticated ways that in the end possibly pacify that user by creating a delegated agency. Linking imaginations and anticipations of future behaviour(s) to categorical renderings from computerized memory, they argue that such an approach "risks delegating whole sets of decisions and, along with that, the ethics and politics of those decisions, to invisible and sentient systems". As they become seamlessly integrated through mobile devices into local, urban environments, such practices "enact and organize global and transactional flows producing an ongoing geography of distanciated, technological performance" (Crang & Graham 2007).

Shush: I'm trying to listen

Many location-based urban interventions claim that distributed, real-time participatory networks can make visible invisible networks of power and responsibility, thus transforming the space of the city into an active agent in the production of new and improved outcomes for public culture and the spaces of the city. But to students of political and historical geography, the environment has always recursively influenced and been influenced by action; this process is not contingent on the possibilities of ambient distributed intelligence. This was the insight of urban sociologists such as Henri Lefebvre and Manuel Castells in the late 1960s and early 1970s, who asserted that the realisation of political goals requires the "production of an appropriate space" (see Pinder 2005).

In the context of these debates, my own research explores listening as a practice attuned to the auditory dimension of technologically-mediated experience. Drawing on historians' attempt to 'hear history', I consider listening in as an alternate methodology for representing place in the networked ecologies of global information environments. This approach foregrounds an embodied, sensual approach to the experience of networked environments against the more dominant visual representational techniques of network mapping, in which places – and people – come to form nodes on a network (Castells, 2000). An ear to the auditory dimension, it is argued, offers an embodied, if tentative, response to some of the 'epistemological traumas' (Boyer, 2006) associated with network mapping.

This listening practice draws on an 'auditory turn' described by Stephen Conner (1997) and others (Thomas, 2007; Smith 2004), which has sought to underscore the importance of the auditory dimension in the cultural and technological constitution of the modern self. This turn towards the auditory dimensions of technological change has itself drawn on phenomenologies of place as articulated by those such as Bachelard (1994) and Palaassma (1996) to enliven the possibility of more sensory, embodied and receptive subjectivities against the "the epistemological regime of the eye" which was accomplished by a separation of the active, transforming self from a nature progressively conceived as passive, constraining and unconscious (Conner 1997, p.203). As Conner writes "the auditory self is an attentive rather than an investigatory self, which takes part in the world rather than taking aim at it" (ibid, p.219).

Listening in to sound history has in turn offered different accounts of the dimensions of technological experience: Alain Corbin's study of 19th century countryside France for example carefully documents the correlations between loudness of bells and the relative significance of various parishes and municipalities (see Bijsterveld 2001, p.45), thus disclosing the associations of technologies of noise or loudness with power. R. Murray Schafer's concept of the 'soundscape' presented the world as a "macrocosmical musical composition", with an ambience of substantially greater decibels accompanying both the Industrial and Electrical revolutions (Schafer, 1977). Historian Peter Bailey has called for greater perception towards the history and meanings of 'noise' as a symbol of sound and its relation to technology (see Bijsterveld 2001, p.39). Emily Thompson has also studied the way architects of the 1930s were able to limit the reverberatory impacts of noise using new building materials, in such a way they created a new, modern interior soundscape. This reconceived reverberation as 'noise' and meant it "lost its traditional meaning as the acoustic signature of a space, and the age old connection between sound and space […] was severed" (Thompson 2004, p.333).

Sound practitioners working in the field of acoustic ecology have progressed a conservationist ethos in the face of such technology changes, seeking to protect endangered sounds from the destructive impacts of relentless technological modernity. Commencing in the late 1960s, R.Murray Schafer (1977) worked with others such as Hildegaard Westerkamp in Vancouver to collect and notate sounds for the *World Soundscape Project*, including police signals, telephone rings, foghorns and sewing machines. This asserted the need to furnish any historical account of the changing auditory dimensions of cities with documentary evidence, what Schafer called ‘soundmarks’, so that listening to the city’s historical changes need not always rely on ‘earwitness’ accounts from literature and mythology.

What happened here?

This paper has explored the way in which listening in to sound history offers some alternative, practice-led methodologies through which to engage with questions about the shifting experience and production of urban spaces. Listening in not only helps frame a more embodied, situated response, but can also provide a more historically-informed understanding of the shaping of spatiality through media events and experiences.

As sound historians of technology have found, the nature of listening has always, to an extent, been influenced by perceptions of noise, and specifically technologies of power and their shaping in different social contexts. Furthermore, excavating the documentary archives locates the role of networked radio in shaping of the spatiality of its listeners. Duckmanton’s VP Day recording was the first of post-war radio reporters who used mobile recording technology to offer his listeners an immediate and direct experience of an event. As Thomas has explained, two years later in 1947 the ABC established a program called *Australian Walkabout* which promised listeners the opportunity to “hear exciting incidents from Australia’s Past and Present”. Walkabout was a term originally applied to the ‘blackfellow’, however it came to articulate ‘an authentic and distinctively Australian mode of engaging with the country’. From the “cool dark comfort of an armchair” listeners could now share with field reporters in the heady rush of discovering the country and of ‘being there’, while in fact being somewhere else (2007, pp.18-20).

Thomas’ analysis of post-documentary radio reportage techniques illustrates the extent to which networked radio helped to shape Australians’ knowledge of their towns and country. This point has also been taken up by Eric Gordon, who has written of the extent to which radio’s emergence as a popular medium helped to shape an ethos of ‘networked urbanism’ which “brought the invisible to the forefront of everyday life and significantly altered how a city could be imagined” (Gordon 2005, p.252). The wonder of radio, in promoting ‘interconnection’ allowed for “connectivity without centralisation” gave rise to new conceptions of city space. Gordon quotes the *New York Times* in 1912:

> Few New Yorkers realize that all through the roar of the big city there are constantly speeding messages between people separated by vast distances, and that over the housetops and even through the walls of buildings and in the very air one breathes are words written by electricity.

As an archeology of the street, this research has also deliberately focused on locations whose transformation over time has reflected shifts in the productive environments of Sydney. Pyrmont, where much of Sydney's sandstone comes from, Victoria St, witness to some of the most controversial developments in Sydney, and witnessing the classic clash between the city as home and as money urban environmental activism, which have not only been noisy affairs and generated media attention, but have also represented times when the street itself has been the subject of dispute.

Listening in to the archives of radio production in Australia in this way offers an alternative reading to that of Bull (2004; 2007), whose ethnographic analysis of iPod culture has explored the nature of urban experience amongst iPod users. To Bull, the personalized music player enables the user to recreate a sense of narrative that overlays or reinscribes journeying in public, and by so doing enables an "auditory re-prioritisation of forms of urban experience" (2007, p.5). To Bull, this process contains a dialectic, which at once promotes an idealized or aetheticised experience of 'public' spaces of the city that conform or mimic the listener's desires, which in turn furthers the absence of meaning attributed to those spaces:

> in iPod culture we have overpowering resources to construct urban spaces to our liking as we move through them, enclosed in our pleasurable and privatized sound bubbles (ibid).

Recognising the centrality of the listening experience in shaping an experience of place, Bull draws on the work of Auge to argue that through auditory technologies of contemporary mobility "all spaces are potentially non-spaces". For iPod users, Bull suggests sound become "a way of perceiving the world" (2007, p.6).

But listening in to cities of sound as they have changed through time, simple demarcations between the aural ecology of urban space and the auditory constitution of technologically-mediated spatiality become more problematic. Emily Thompson notes the extent to which noise came to predominate modern cities has necessitated a shift toward 'quality of life' as a factor in noise control, observing that "as consumers in search of a quality soundscape, we are left to take matters into our own hands, and in those hands one finds a gleaming iPod" (2005, p.198). Thompson's remarks suggest that while iPod culture may evidence a tuning of the world according to a mediated listening experience, the extent to which unplugging the headphones somehow facilitates a *less* technologically-mediated auditory experience is questionable. This point has been taken up by Beer (2007) who considers the use of mobile music devices an "integrated yet distracted part of the aural ecology and informational structures of the city".

Within the technologically-mediated listening contexts of today's iPod culture, this listening practice seeks to uncover the buried histories of the media-generated urban soundscape. Taking into account the discoveries of sound historians such as Thompson (2005) and Bijsterveld (2008), the urban context can be listened to as a radical technological artefact, one whose acoustic temperament has been shaped by technologies of power, from church bells to air planes and automobiles. Where Bull's iPod users 'project their desires' onto the spaces of the city, my listening practice asks – a bit playfully – can I use my iPod as though it was a homing device for the history of place? Can I tune in to the residual resonances of this city's development and transformation over time, finding its invisible traces in the networked media of our collective memories?

An exploratory listening experience of the city's history follows a speculative desire to imagine – and hear – cities differently. As put by Elizabeth Wilson (1991, p. 11):

> perhaps we should be happier in our cities were we to respond to them as nature or dreams; as objects of exploration, investigation, and interpretation, settings for voyages of discovery. The 'discourse' that has shaped our cities – the utilitarian plans of experts whose goal was social engineering – has limited our vision and almost destroyed our cities.

Conclusion

Research toward the production of Sydney Sidetracks has identified a place for sound archives within the developing conventions of location-aware mobile computing. This paper has explored the qualities that listening in to the sounds of Sydney's development history can bring to the way spatial contexts are represented through the networked environments of mobile media. This recognises the specific qualities of the auditory dimension as an important but sometimes overlooked aspect of urban experience. Drawing on the insights of sound historians and practitioners attuned to the historical shaping of the modern soundscape also allows for a greater porosity in our approach to contemporary mobile media in urban contexts, one that builds connections not only between the intimate sounds of an iPod's playlist but the aural ecology of technological modernity, and the historical contingencies of our place in the world.

Note: individual audio recordings discussed in this paper can be accessed at http://www.sitesandsounds.net.au or through *ABC Sydney Sidetracks* at http://www.abc.net.au/sidetracks which offers access to recordings online or through a mobile application which can be downloaded direct to the phone.

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